Motivation

Recent advances

Instance segmentation

https://github.com/IDEA-Research/Grounded-Segment-Anything

Panoptic segmentation

https://github.com/IDEA-Research/Grounded-Segment-Anything

Manipulation

https://github.com/IDEA-Research/Grounded-Segment-Anything

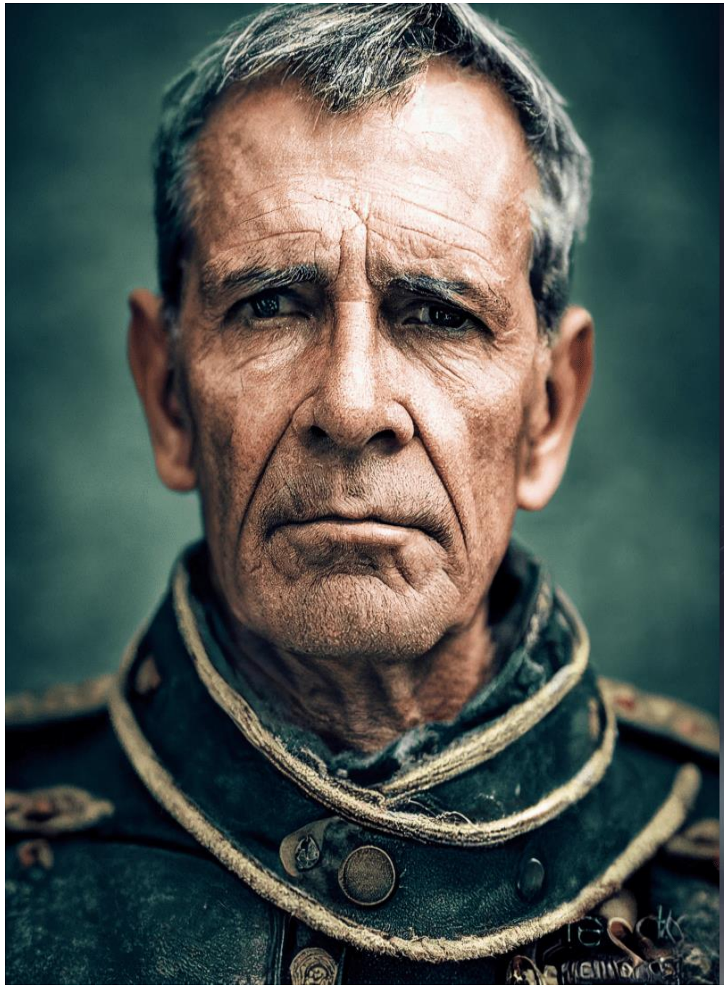
Mesh recovery

https://github.com/IDEA-Research/OSX

3d scene

https://www.matthewtancik.com/nerf

Generative AI



https://stablediffusionweb.com/ https://www.jousefmurad.com/ai/a-primer-on-stable-diffusion/

Questions?

CAP5415
Computer Vision

Yogesh S Rawat

yogesh@ucf.edu

HEC-241

Introduction to Computer Vision

Lecture 1

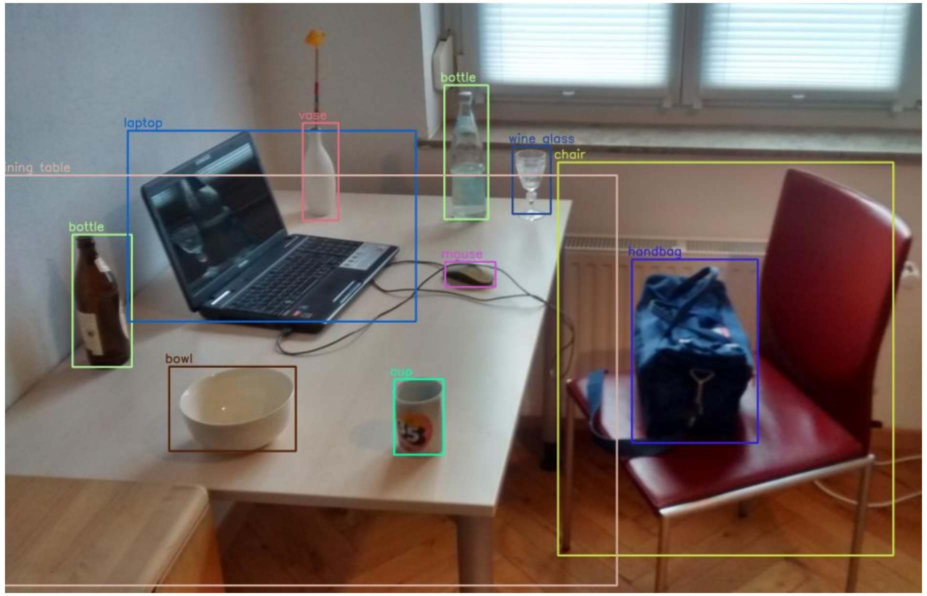
Introduction to Computer Vision

Lecture 1

Basics

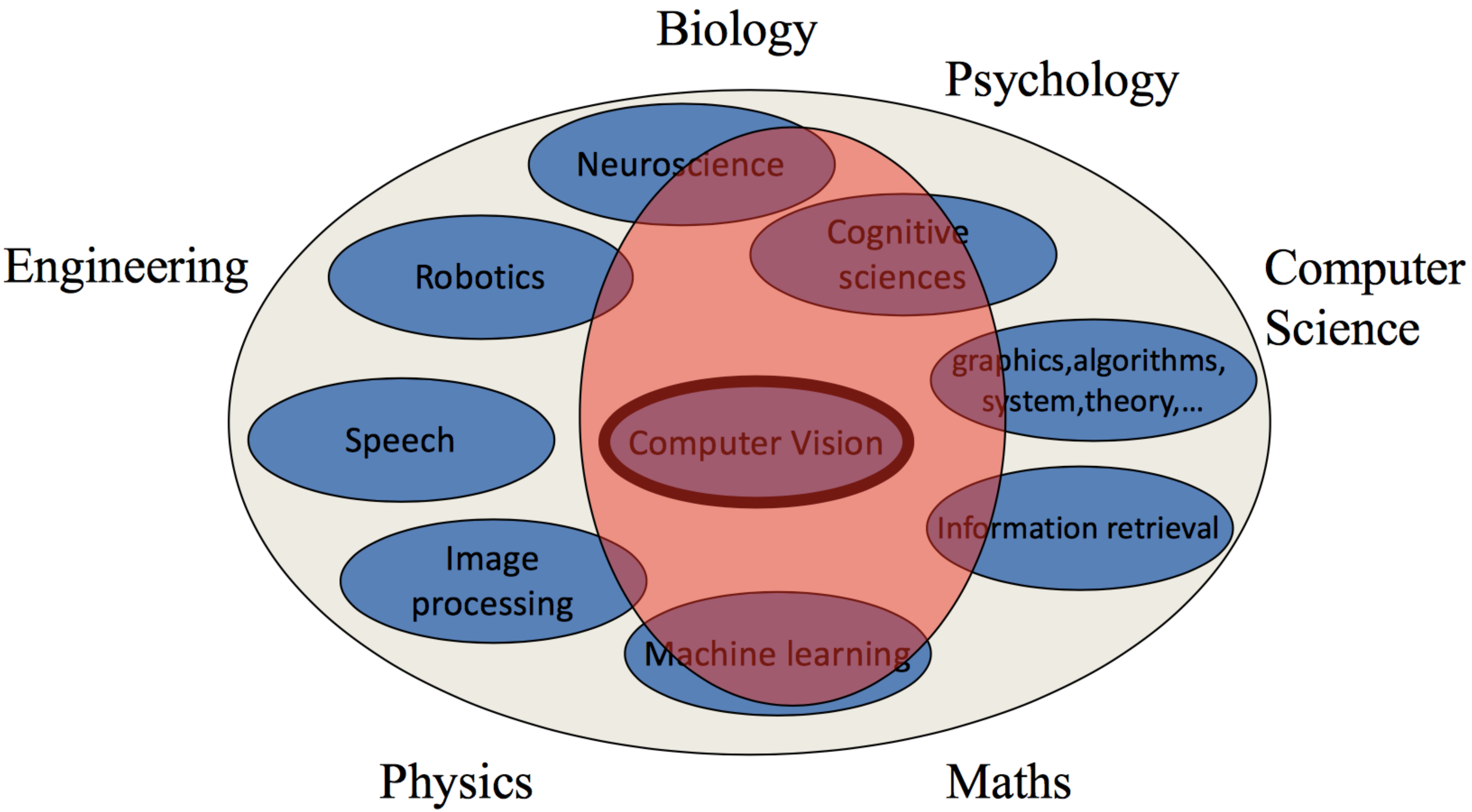
What is computer vision?

- Ability of computers
  - To understand visual data
    - For example, images, videos…



- Automate tasks
  - Which human visual system can perform

What is it related to?



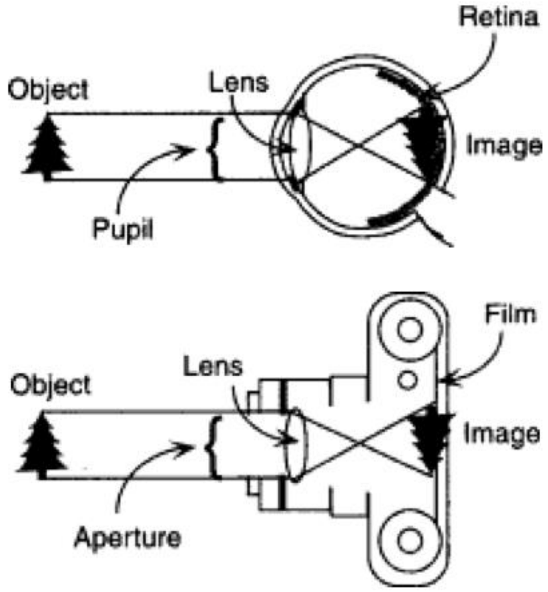
Visual Perception

- How do we know that the objects that we see are for real?
- Can people “see” without being *aware* of what they see?
- Why do objects appear colored?

Visual Perception

- Definition: *Process of acquiring knowledge about environmental objects and events by extracting information from the light they emit or reflect [Palmer, 2012].*

Vision → Cognitive Activity → Acquisition of knowledge



Perception is analogous to taking a picture!
(credit: Palmer, 2012)

Visual Perception

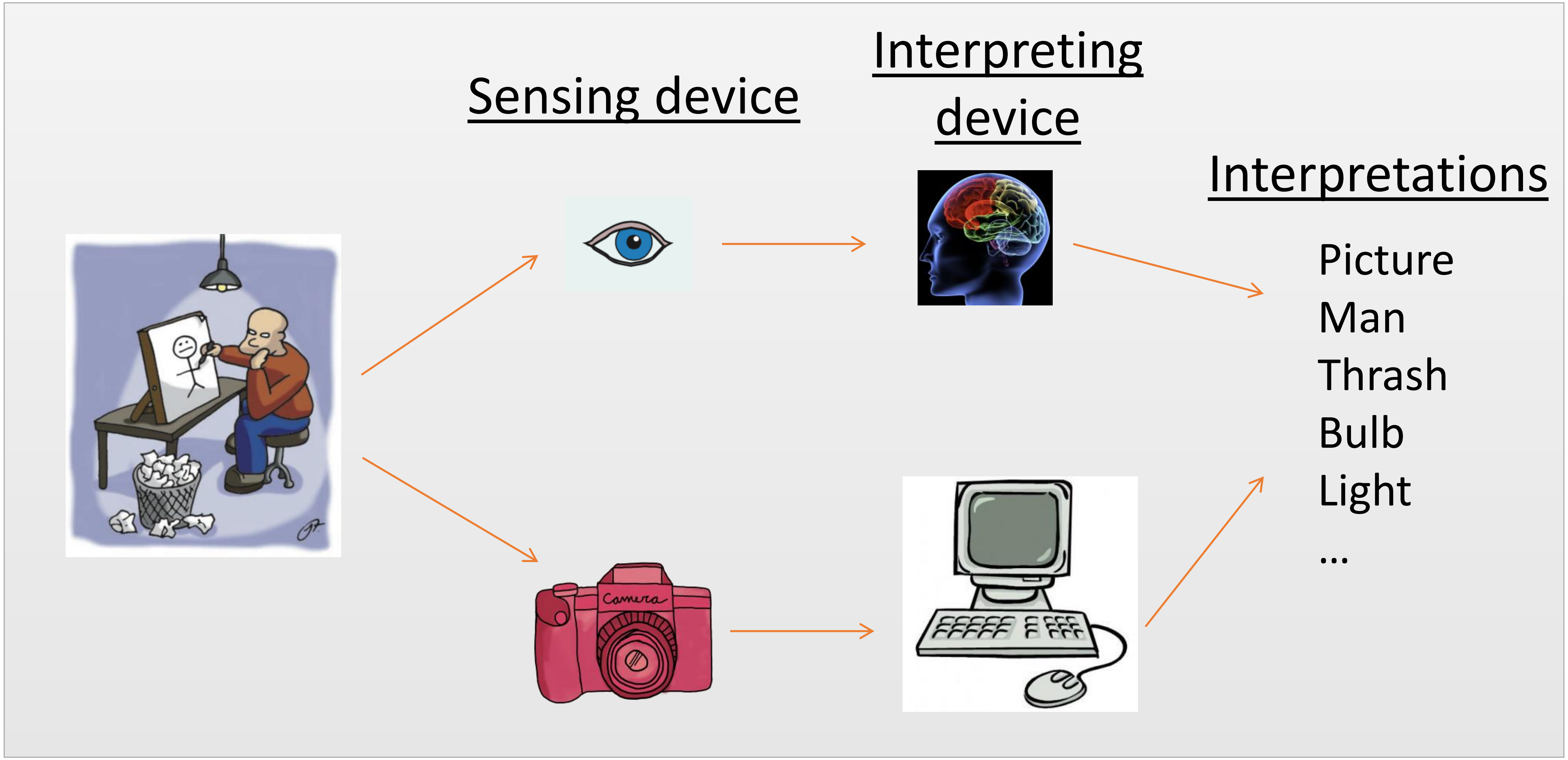
Vision is a process in which temporally changing intensity and color values in the image plane have to be interpreted as processes in the real world that happen in 3D space over time

What can you see in this picture?

Now can you see what the picture is?

Credit: Thompson, Basic Vision, Oxford Press, 2012.

Vision vs. Computer Vision?

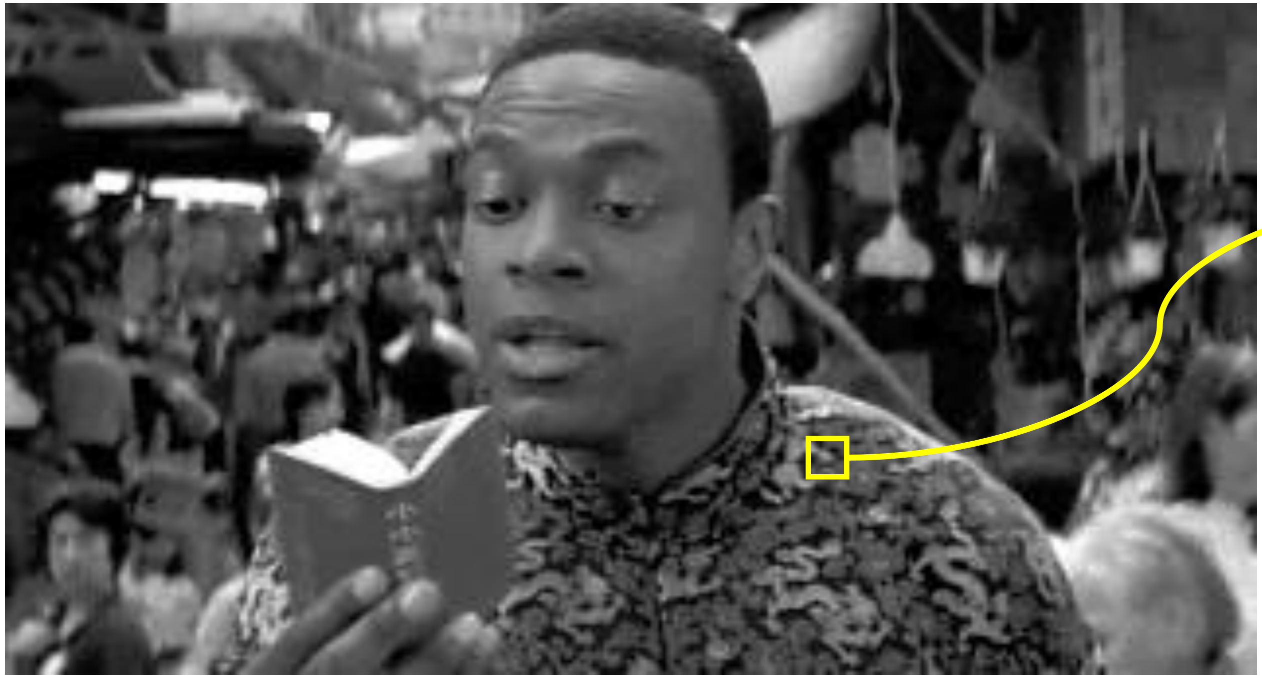


Vision and Image Understanding

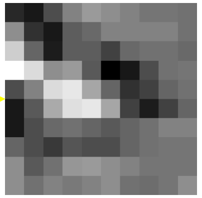
- Visual tasks: We use vision to interact with environments and survive
    - to navigate and avoid obstacles
    - to recognize and pick up objects
    - to identify food and danger
    - friends and enemies
    - …

Goal of Computer Vision?

- To bridge the gap between
  - image pixels and “meaning” (semantic)!



What we see!

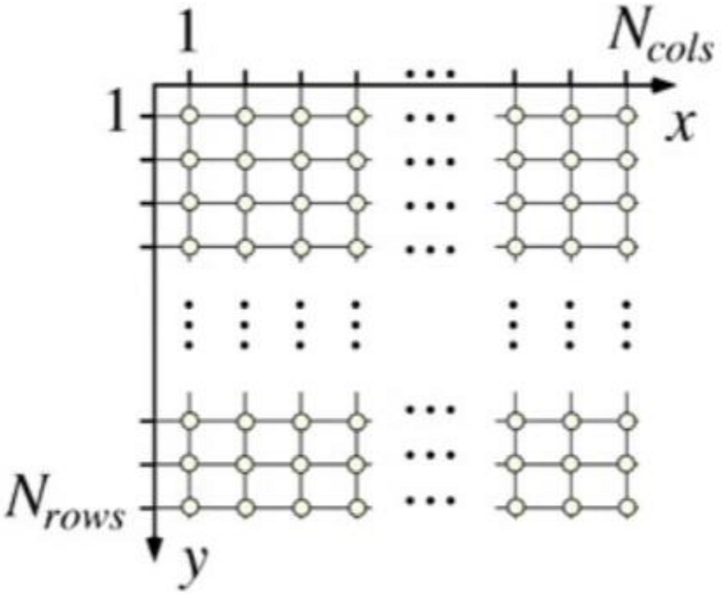


0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What computer sees!

What is a (digital) Image?

- Definition: A digital image is defined by *integrating* and *sampling* continuous (analog) data in a spatial domain [Klette, 2014].



Left hand coordinate system

Image Types: (Gray)Scalar and Binary

- A scalar image has integer values

$$u \in \{0, 1, ..., 2^a - 1\}$$

- a: level (bit)
- Ex. If 8 bit (a=8)
  - image spans from 0 to 255
  - 0 black and 255 white
- Ex. If 1 bit (a=1)
  - it is binary image
  - 0 and 1 only

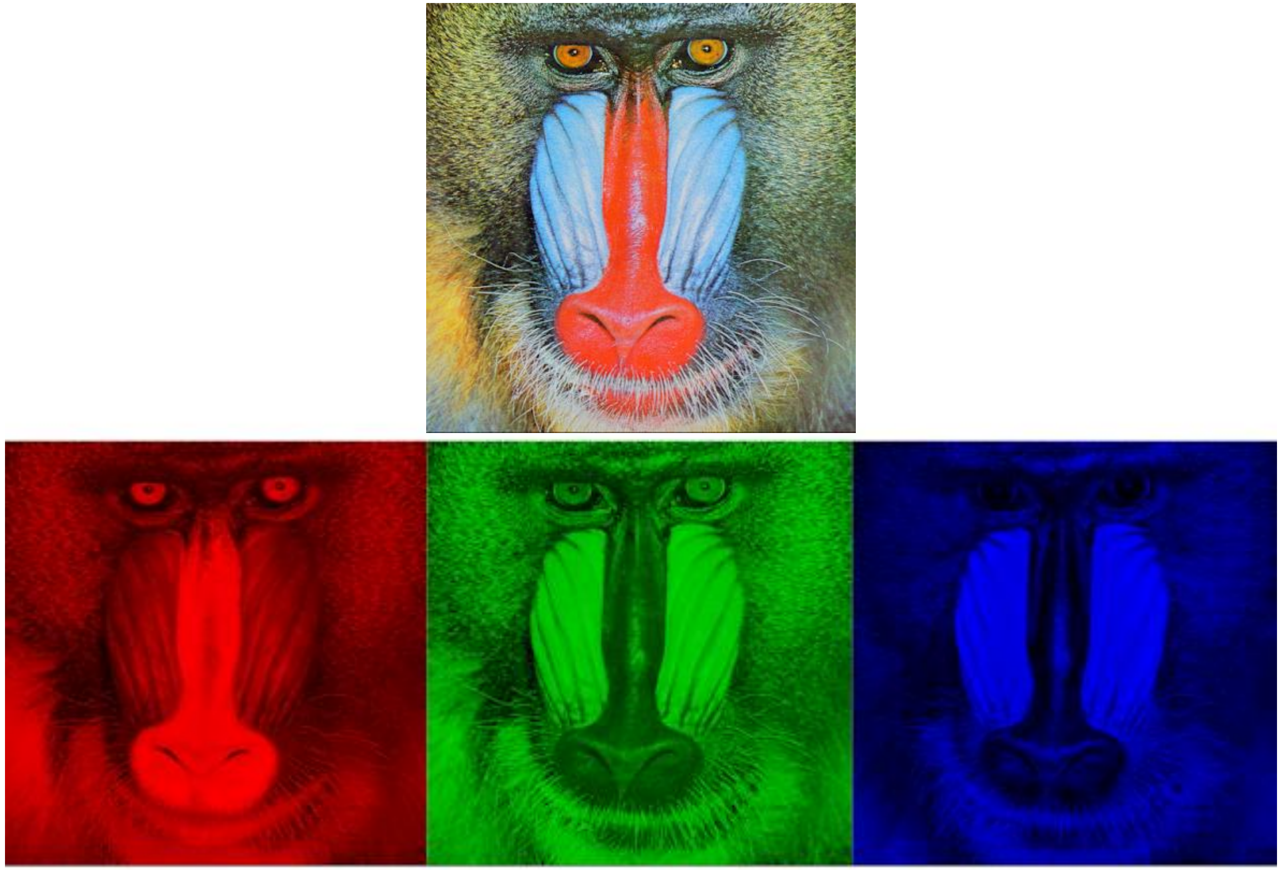
Original
(256 colors)

8 colors

4 colors

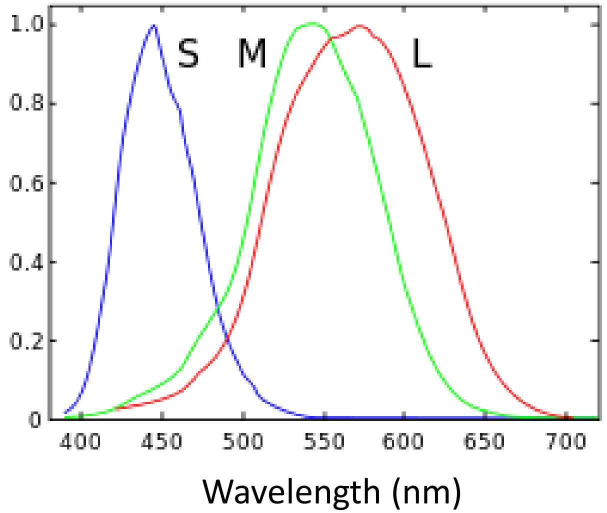
Image Type: RGB (red, green, blue)

- Image has three channels (bands)
- Each channel spans a-bit values.



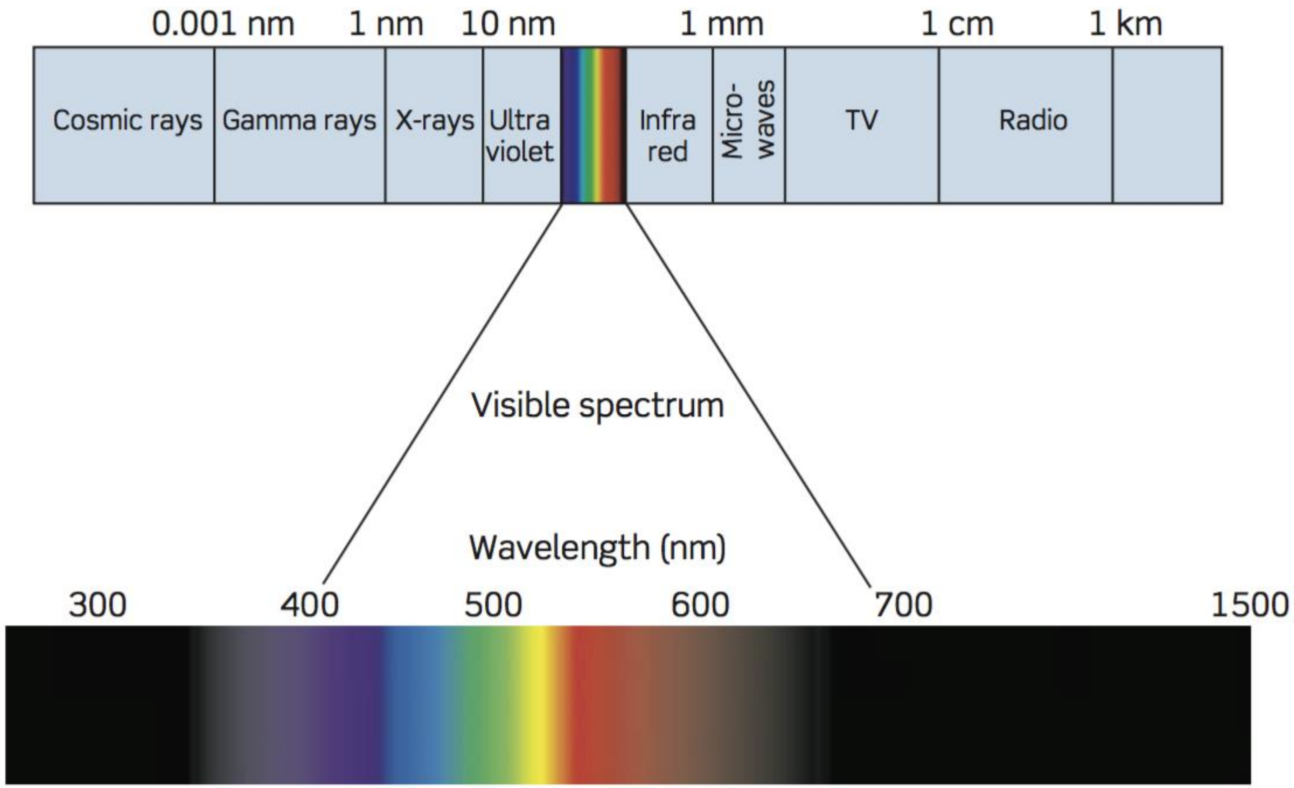
Human Cone-cells (normalized) responsivity spectra

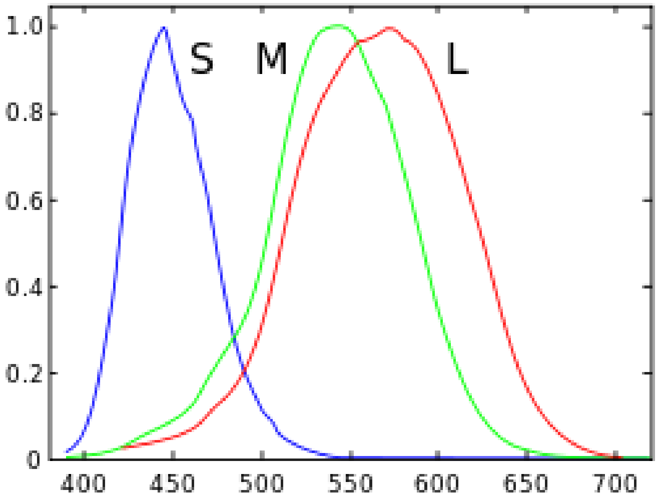
- Some people might have 4 cone-types!
- Some might have just 2!



Color

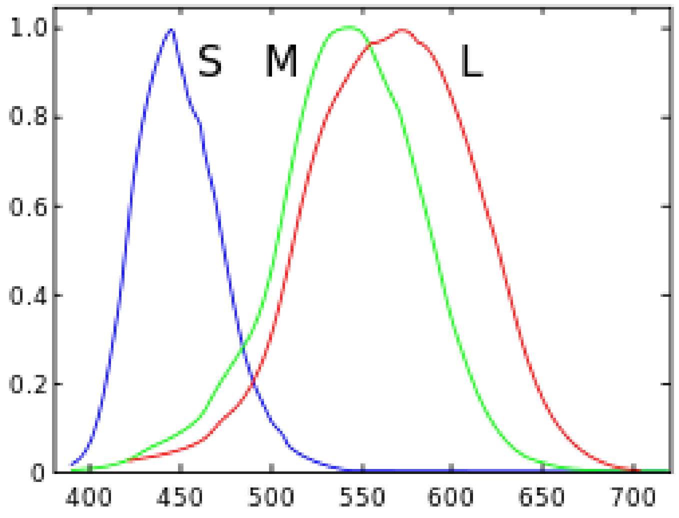
- Color vision has evolved over millions of years.



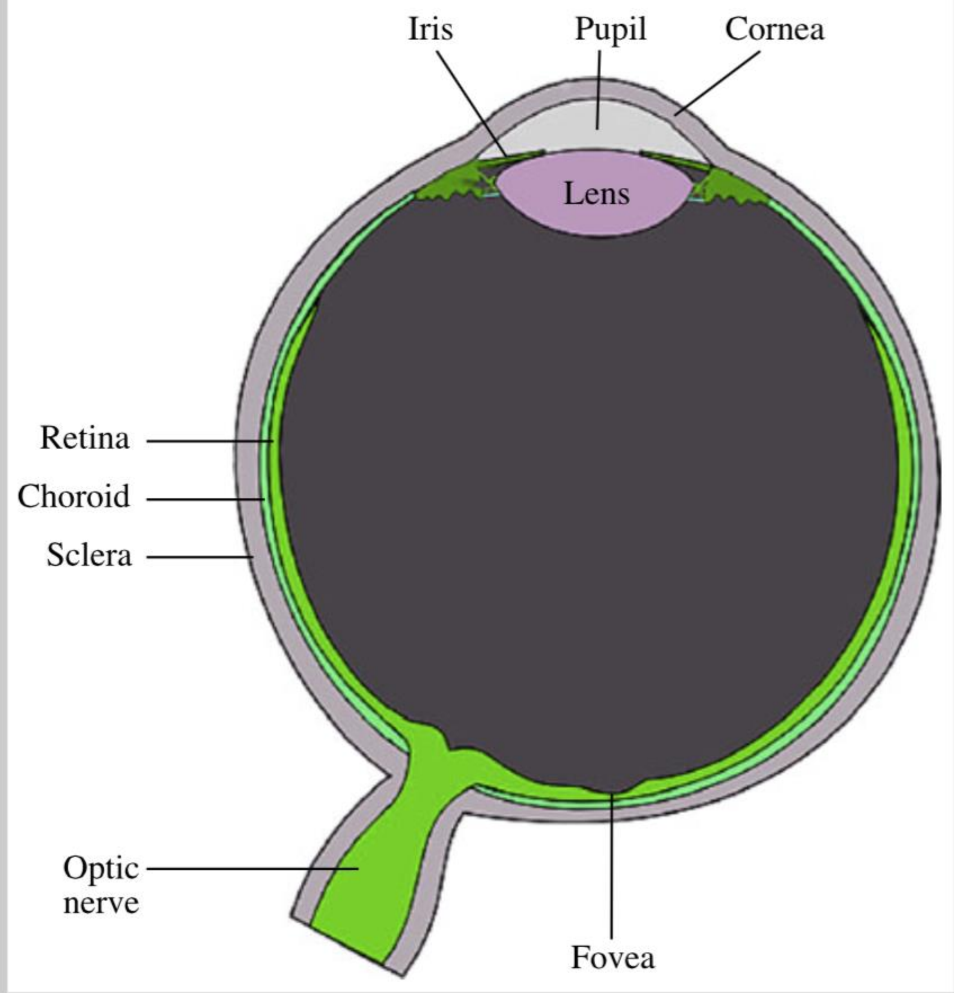


Color

- If there is no light, there is no color!
- Human vision can only discriminate a few dozens of grey levels
  - But hundreds of thousands of different colors.
  - RED -> ~625 to 780 nm [long wavelength]
  - ORANGE -> ~ 590 to 625 nm [long wavelength]
  - YELLOW -> ~565 to 590 nm [middle range wavelength]
  - GREEN -> ~ 500 to 565 nm [middle range wavelength]
  - CYAN -> ~485 to 500 nm [middle range wavelength]
  - BLUE -> ~440 to 485 nm [short wavelength]
  - VIOLET -> ~330 to 440 nm [very short wavelength]



Retina of Human Eye



There are three different types of color-sensitive cones corresponding to (roughly)

- RED (64% of the cones)
- GREEN (about 32%), and
- BLUE (about 2%).

6-7 million cones [for colors]
120 million rods [for light]

Some may have only 2 cones
Dogs also have 2 types

Credit: Klette, 2012.

Questions?

Introduction to Computer Vision

Lecture 1

Motivation

Motivation

Why study computer vision?

Because…

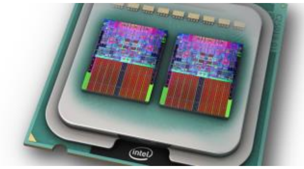
- Find it interesting!
- Out of curiosity!
- Course requirement!
- Good job market!

Recent advancement

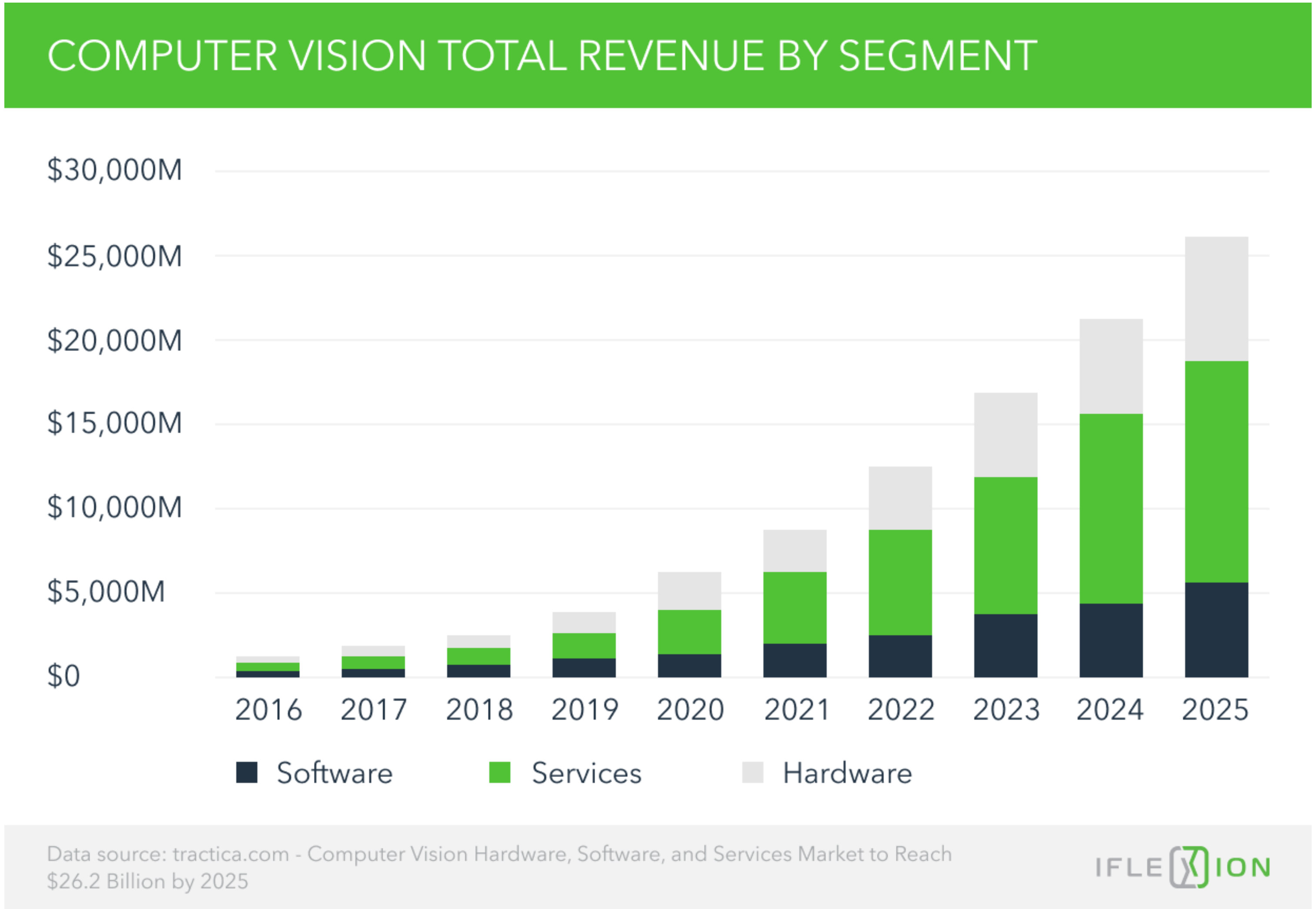
- Used to be done mostly in academics
- Now
  - Google
  - Facebook
  - Apple
  - IBM
  - Microsoft
  - …

What changed?

- Emergence of deep learning
- Advancement in hardware
- Availability of large-scale data
    - ImageNet
    - OpenImages
    - YFCC100M
    - Youtube-8M
    - Kinetics
    - AVA
    - …



Market of computer vision



Source: https://www.iflexion.com/blog/computer-vision-applications

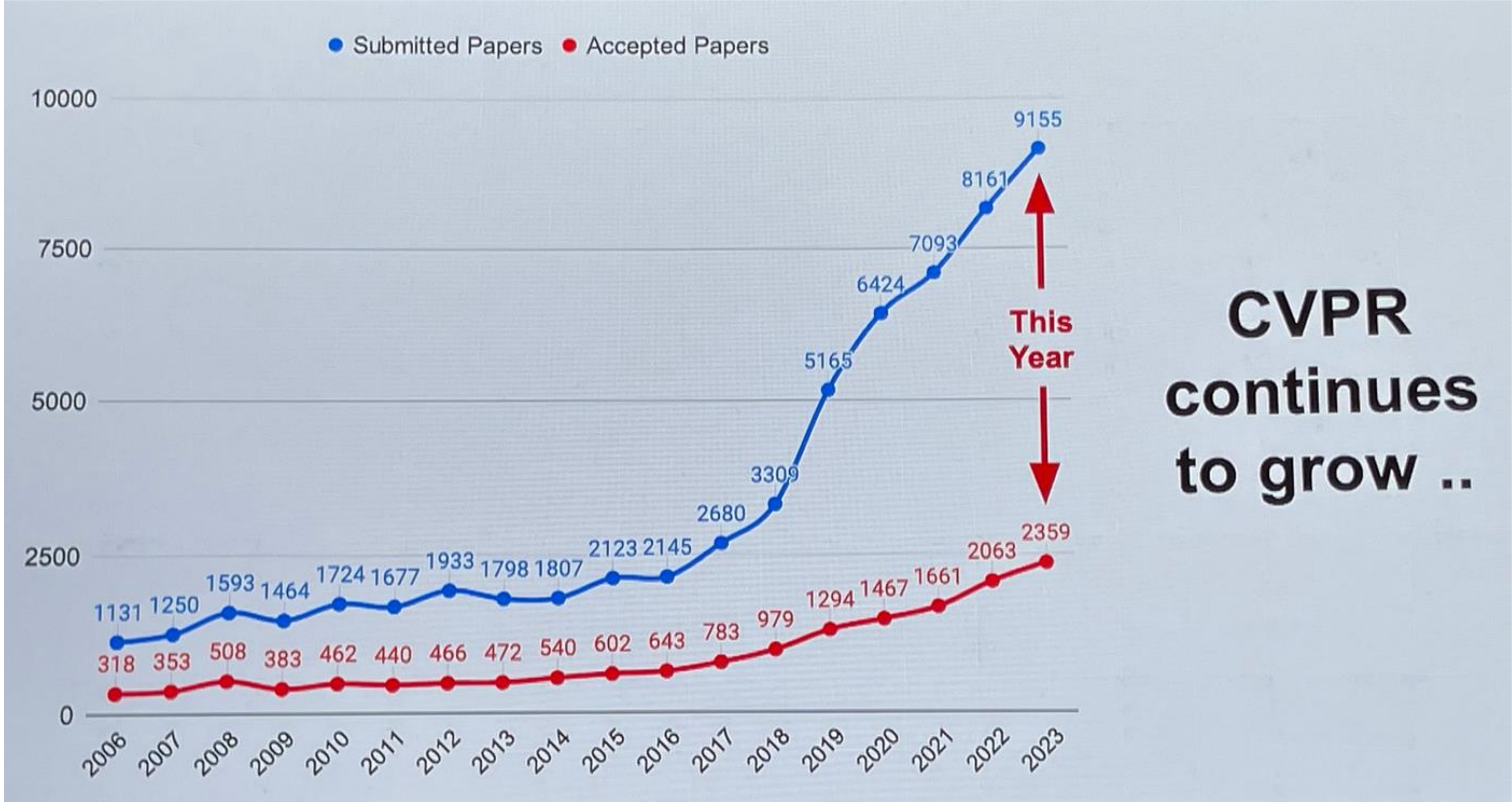
CVPR conference ranking (Engineering)

	Publication	h5-index	h5-median
1.	IEEE/CVF Conference on Computer Vision and Pattern Recognition	389	627
2.	Advanced Materials	312	418
3.	International Conference on Learning Representations	286	533
4.	Neural Information Processing Systems	278	436

CVPR conference ranking

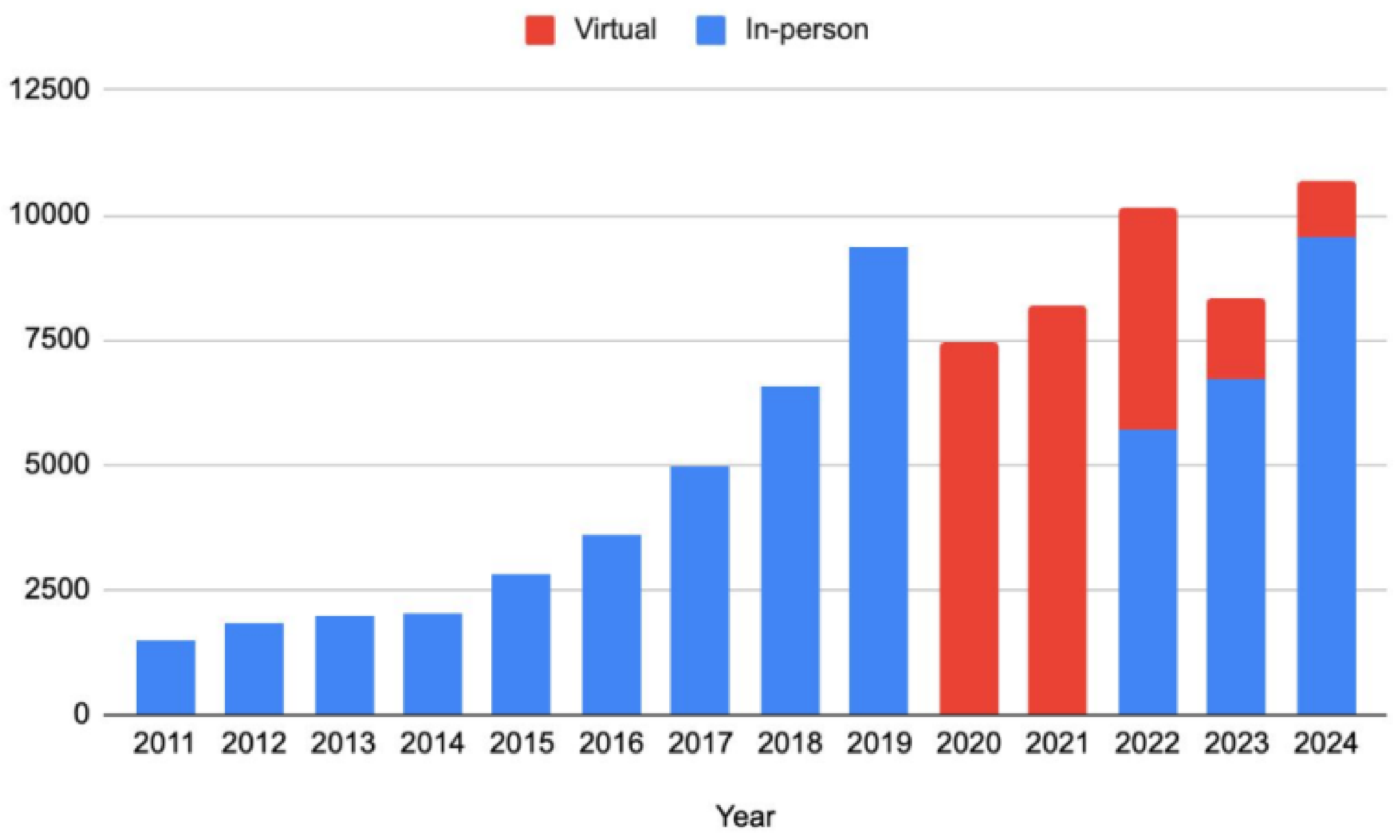
	Publication	h5-index	h5-median
1.	Nature	444	667
2.	The New England Journal of Medicine	432	780
3.	Science	401	614
4.	IEEE/CVF Conference on Computer Vision and Pattern Recognition	389	627

CVPR - papers



https://medium.com/@dobko_m/cvpr-2023-summary-ad271d383404

CVPR attendance



*As of 6/12

Questions?

Introduction to Computer Vision

Lecture 1

Applications

Applications

Retail – Amazon Go

https://youtu.be/NrmMk1Myrxc

Retail - Clothing

https://youtu.be/Mr71jrkzWq8

Self-driving - Waymo

https://youtu.be/TsaES--OTzM

Autopilot - Tesla

https://youtu.be/UgNhYGAgmZo

Questions?

Introduction to Computer Vision

Lecture 1

Tasks

Computer vision tasks

Object Recognition

- Problem: Given an image A, does A contain an image of a person?

Object Recognition

- **Problem:** Given an image A, does A contain an image of a person?

Phil Noble / AP

Sydney Morning Herald

Patrick Gardin / AP

Andy Barron / Reno Gazette-Journal

YES

CENTER FOR RESEARCH IN COMPUTER VISION

Object Recognition

- Problem: Given an image A, does A contain an image of a person?

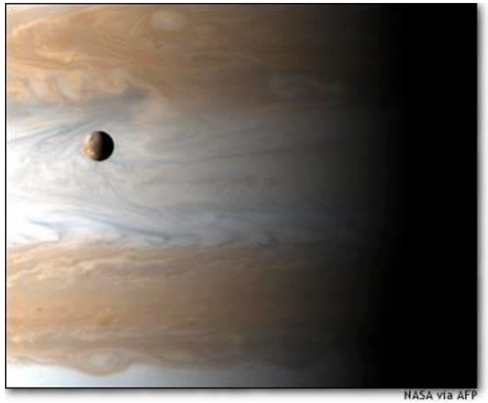
Monroe County Sheriffs Department / Newsmakers

Mark Garkfinkel / The Boston Herald

Jeff J. Mitchell / Reuters

NO

Uno Andersson / AP



NASA via AFP

Object localization

Phil Noble / AP

Mike Hewitt / Allsport

Patrick Gardin / AP

Andy Barron / Reno Gazette-Journal

Sydney Morning Herald

Human Detection

Semantic Segmentation

Image source: Antonio Torralba

Segmentation

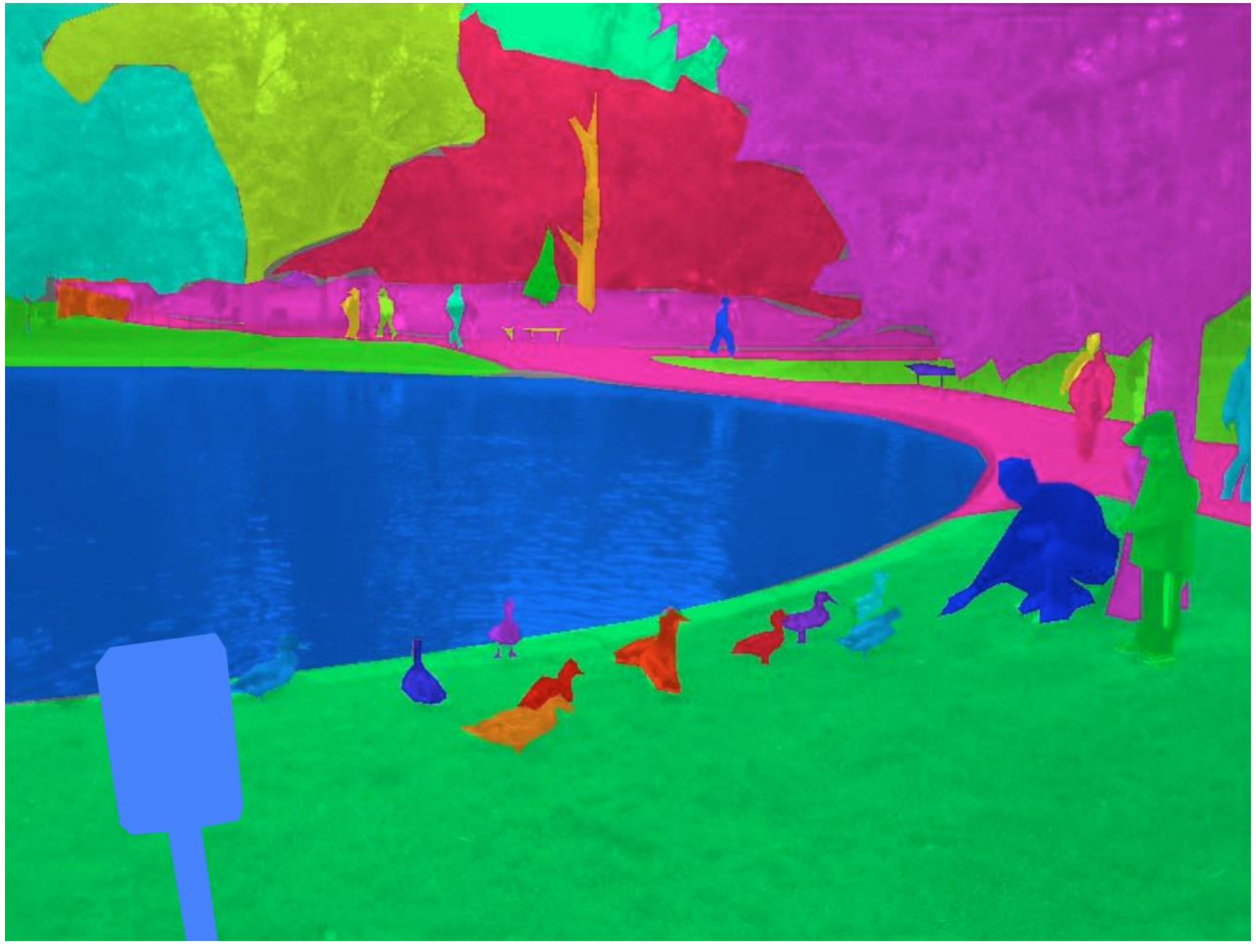
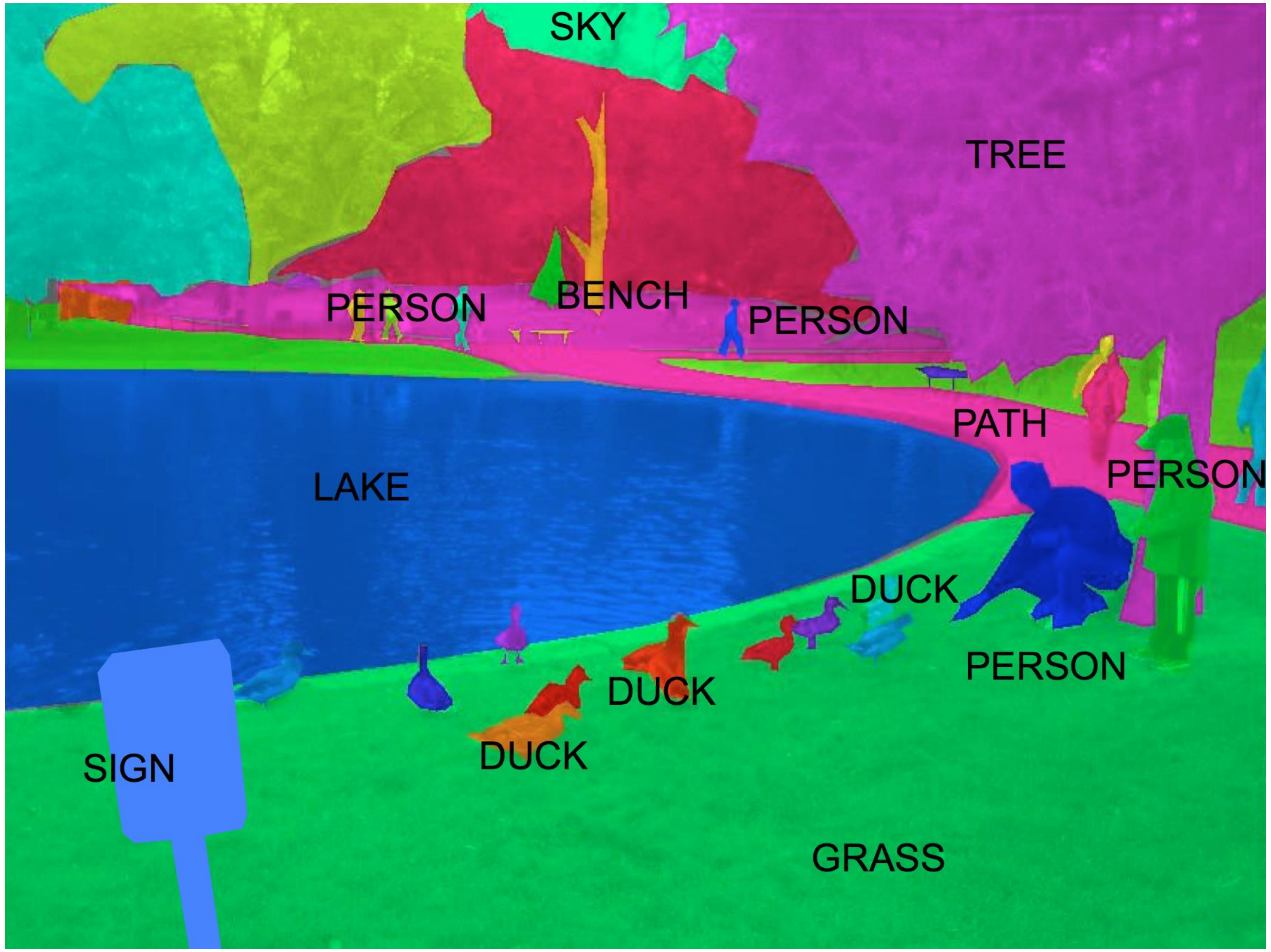


Image source: Antonio Torralba

Segmentation



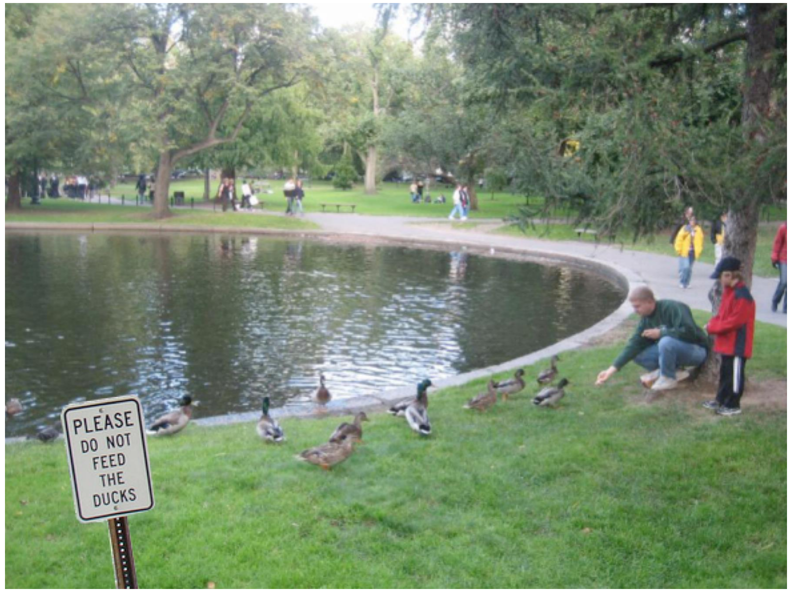


Image source: Antonio Torralba

Segmentation

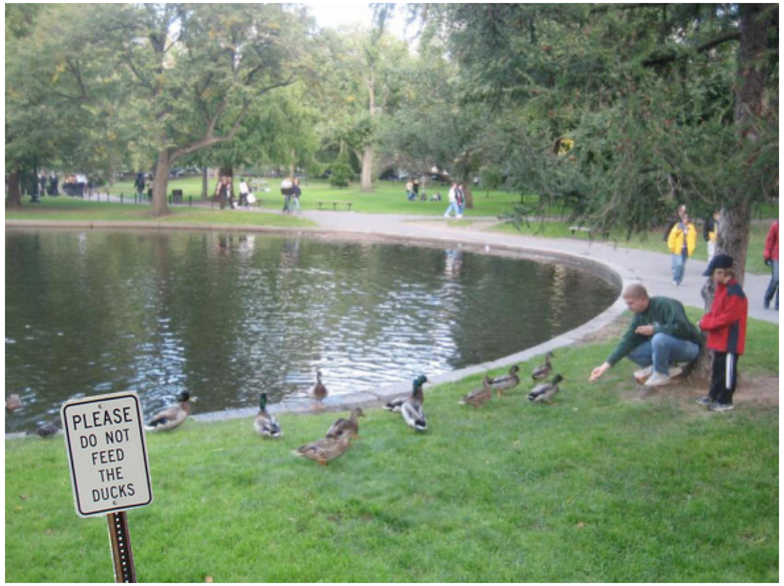


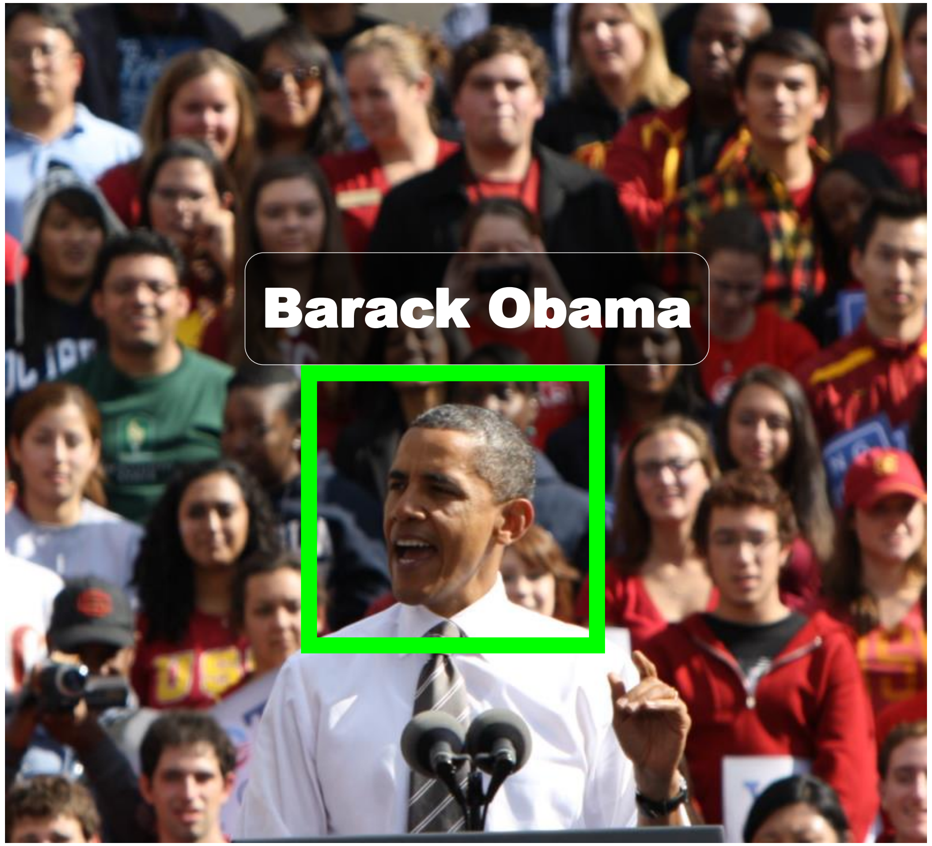
Image source: Antonio Torralba

Semantic part labeling

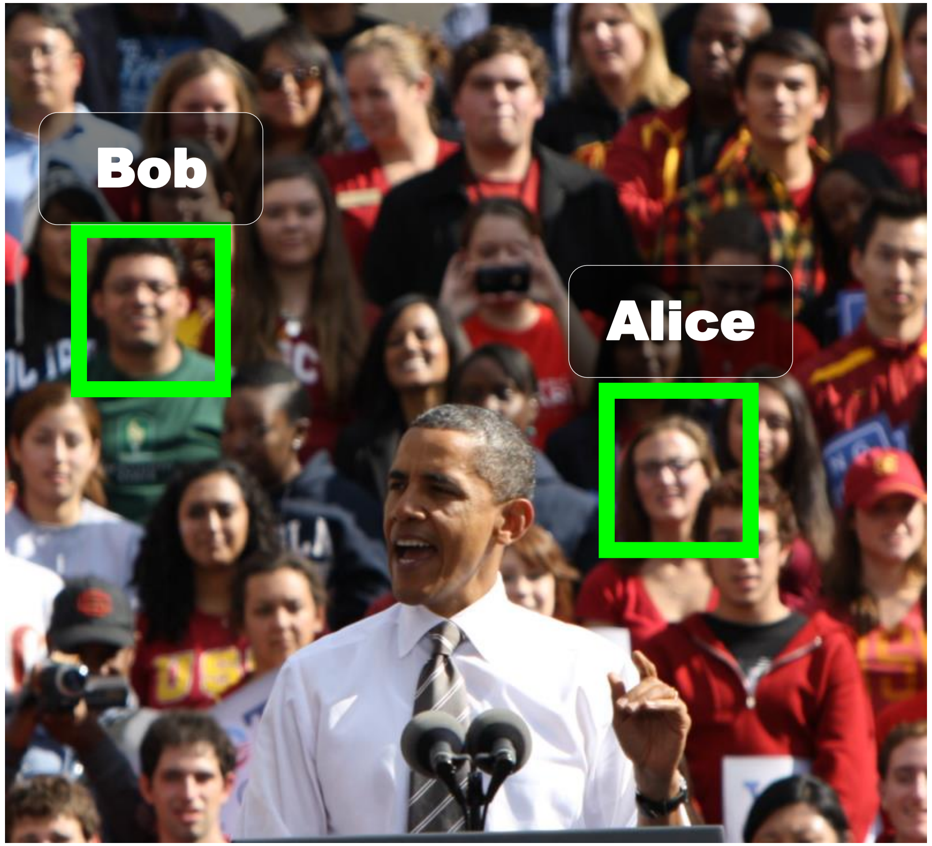
Face Recognition

Open-Universe Face Identification

News Article: Label Important Figures



Social Network: Tag Facebook Friends

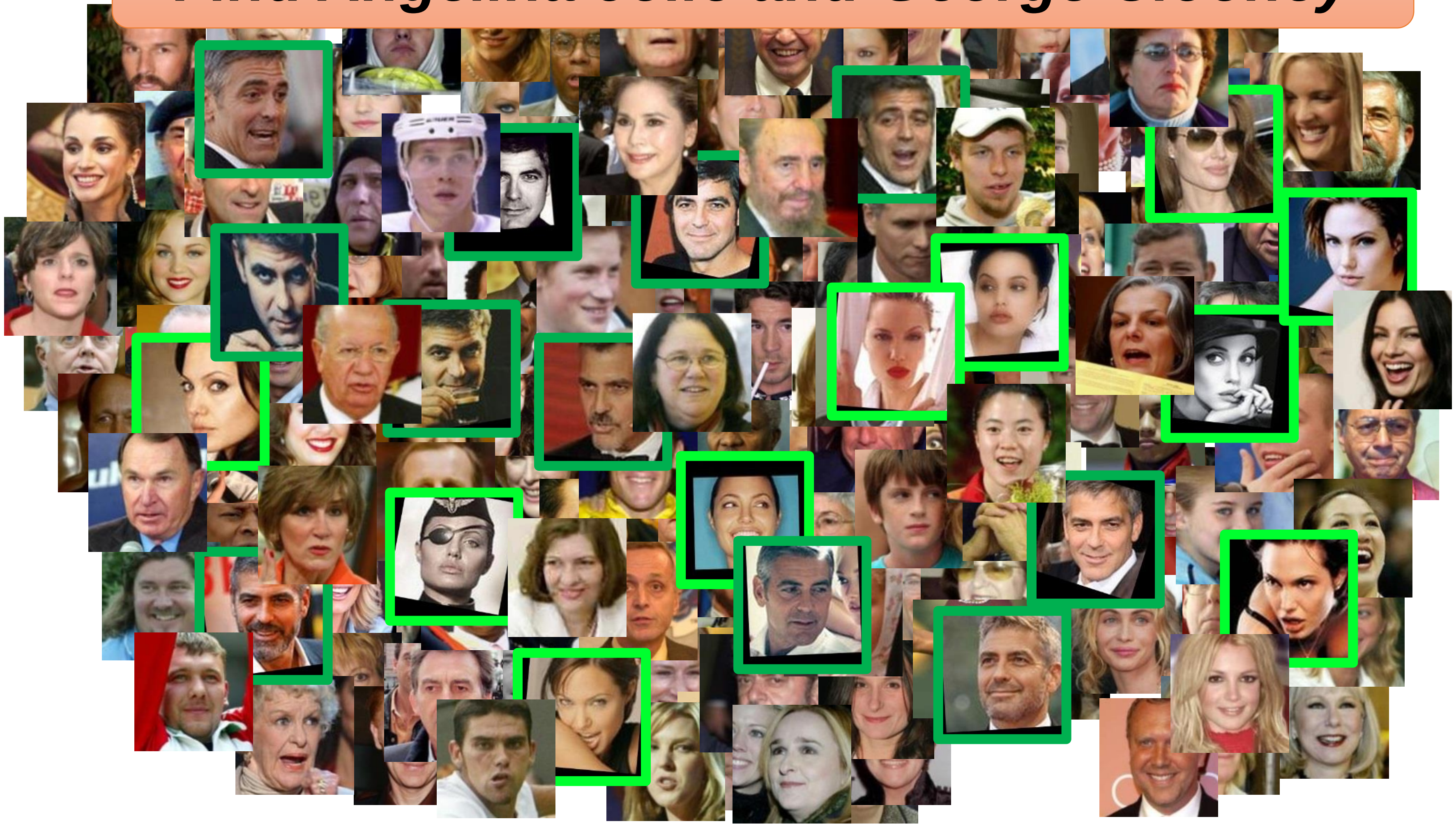


Open-Universe Face Identification

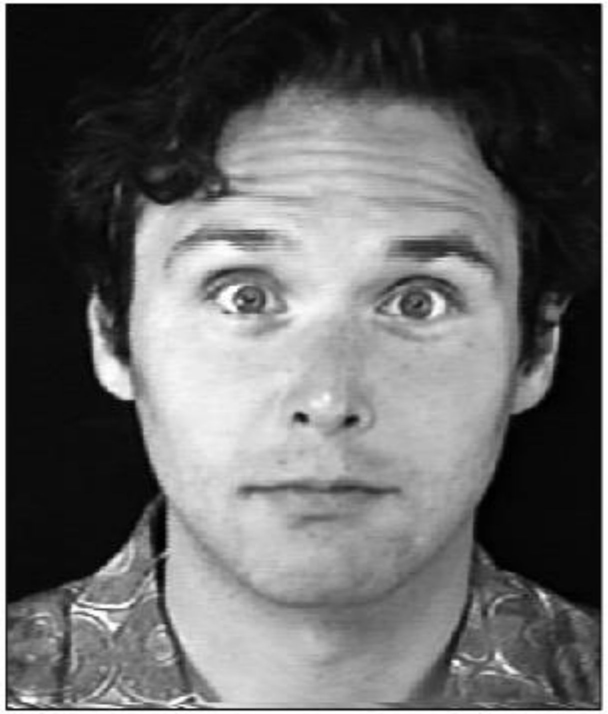
Find Angelina Jolie and George Clooney

Open-Universe Face Identification

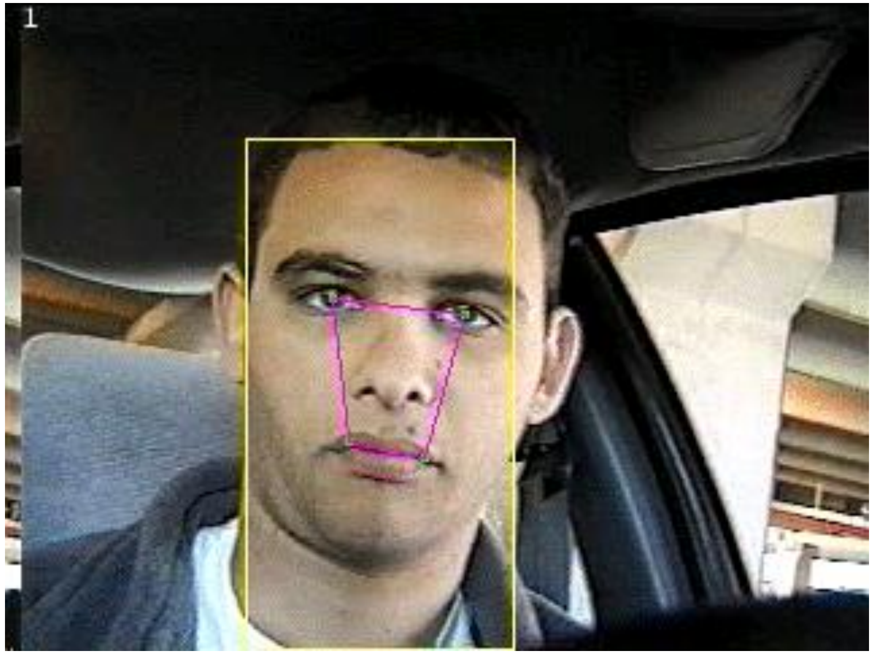
Find Angelina Jolie and George Clooney

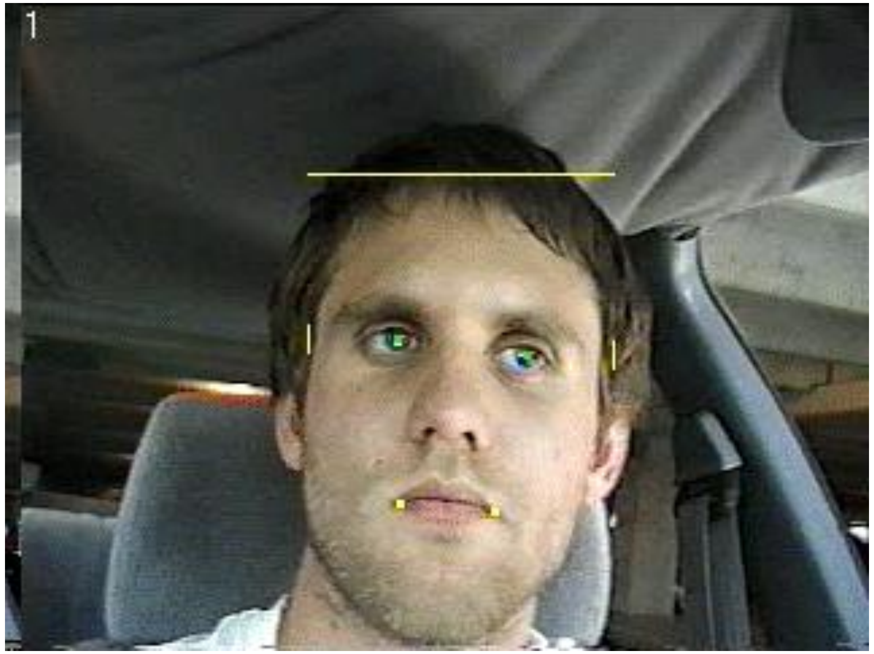


Facial expression

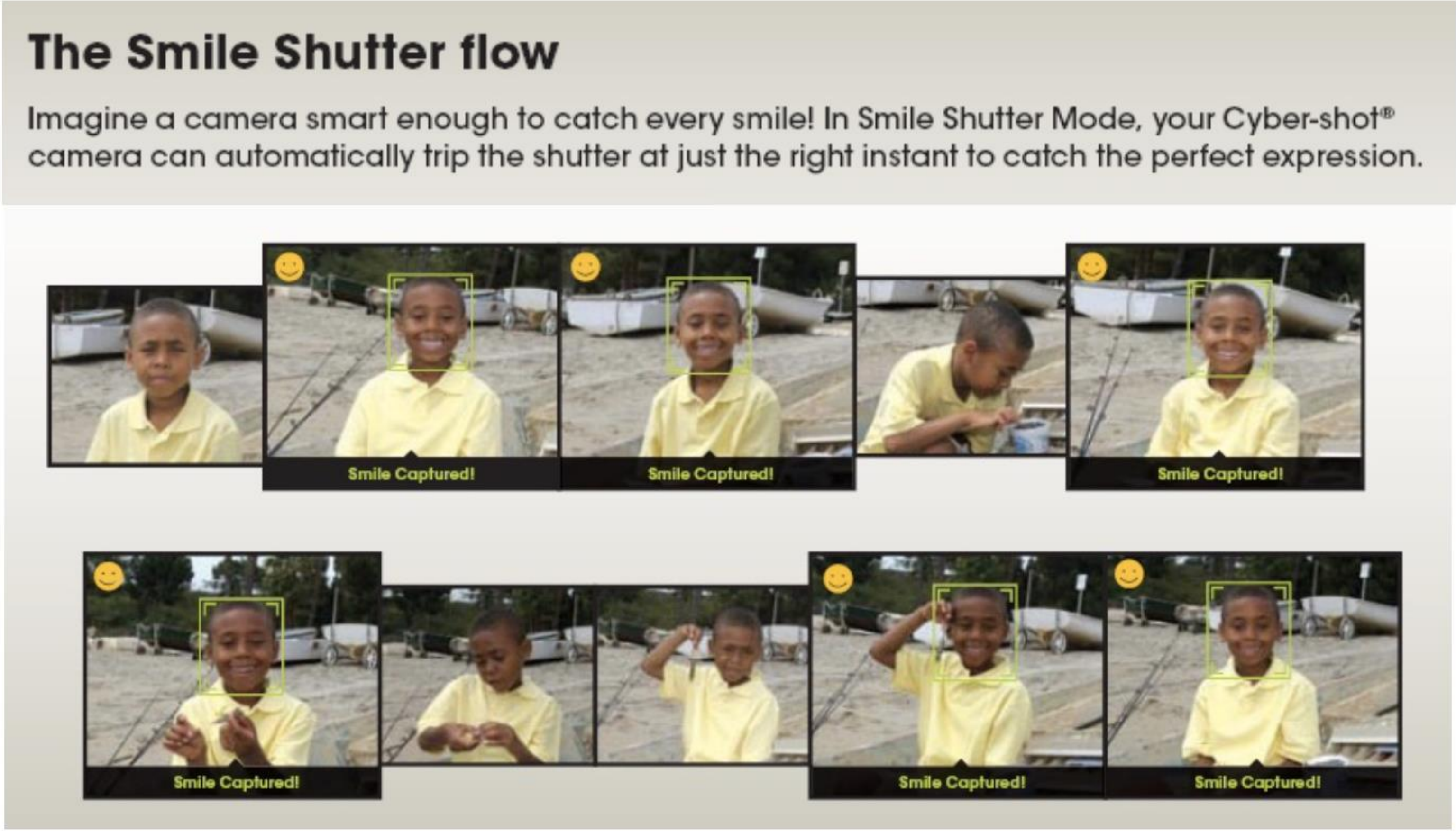


Fatigue detection





Smile detection



[Sony Cyber-shot® T70 Digital Still Camera]

Source: S. Seitz

High Density Crowded Scenes

Political Rallies

Religious Festivals

Marathons

High Density
Moving Objects

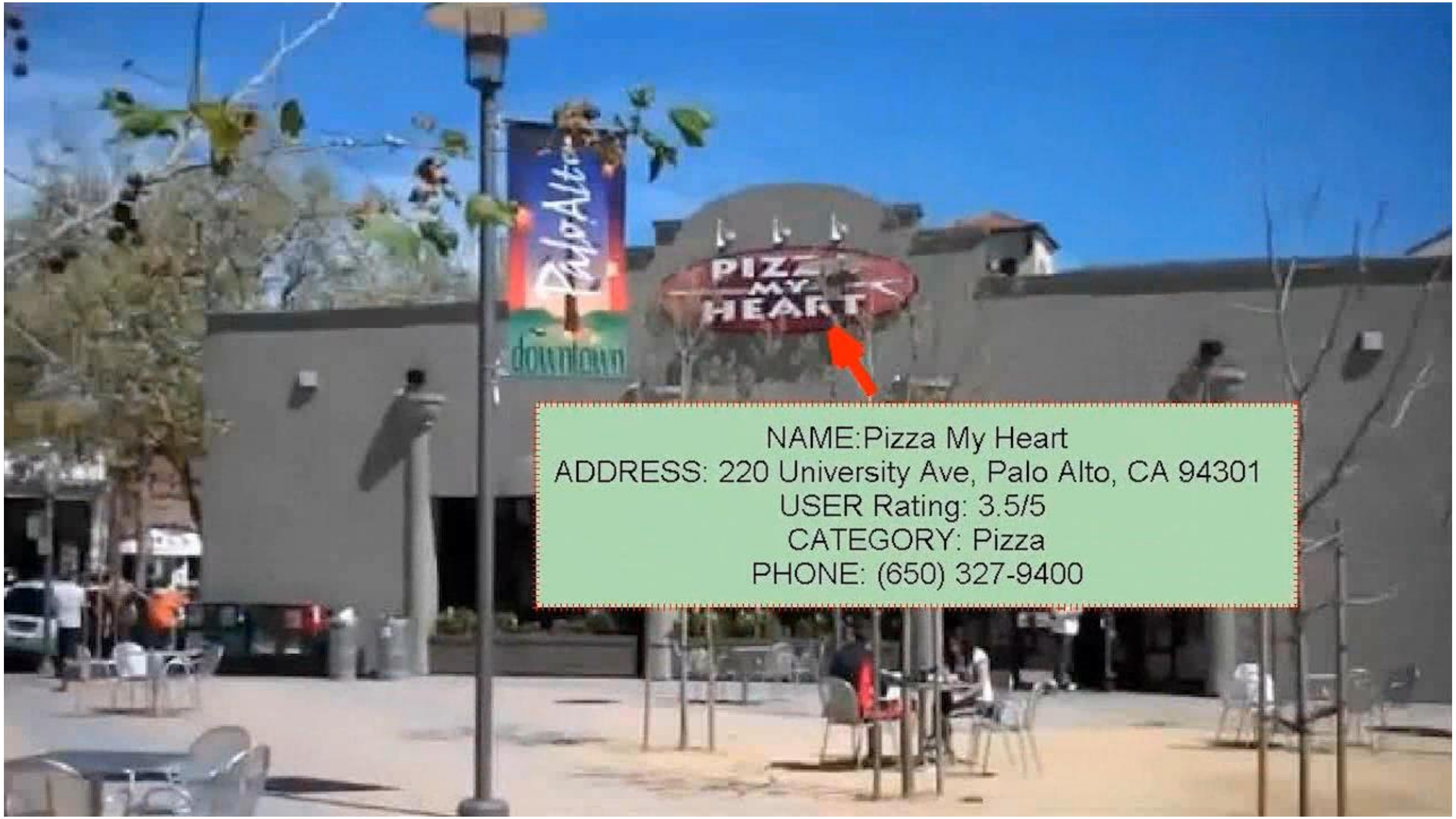
Counting

Ground truth=634 Proposed Method by Idrees and Shah=640

Counting

Ground truth=2319 Proposed Method=2496

Visual Business Recognition



Video Clip

- Sequences of frames
- 30 frames per second

Sequences of Images

Action recognition – UCF101

Cycling

Diving

Golf Swinging

Riding

Volleyball

Basketball Shooting

Swinging

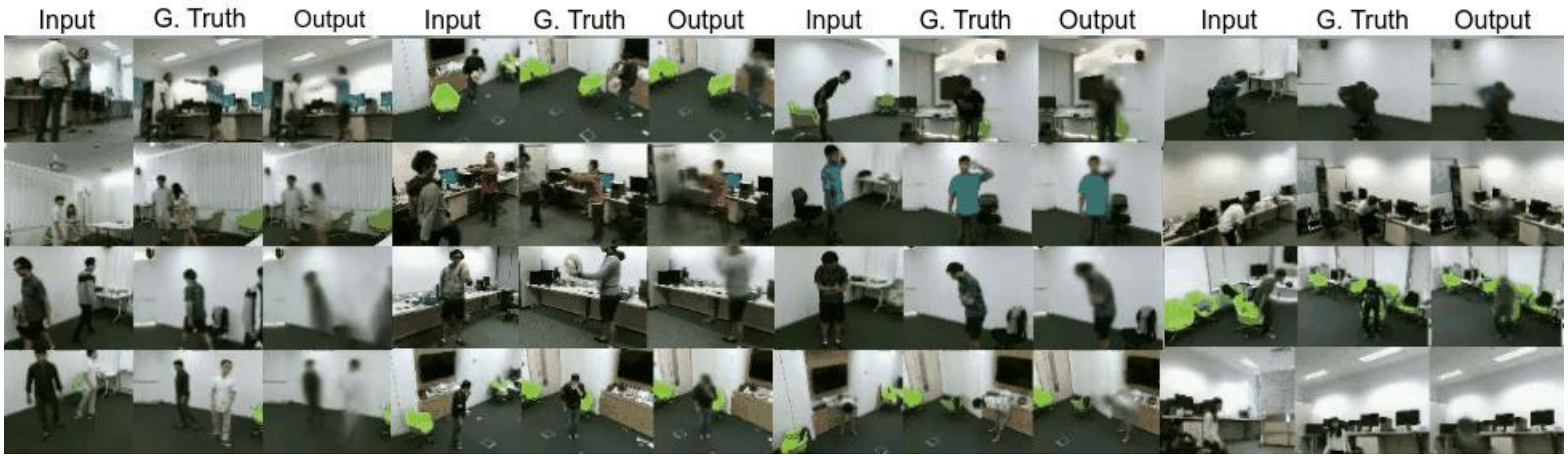
Tennis Swinging

Action detection

Video segmentation

Duarte, Rawat, Shah, ICCV 2019

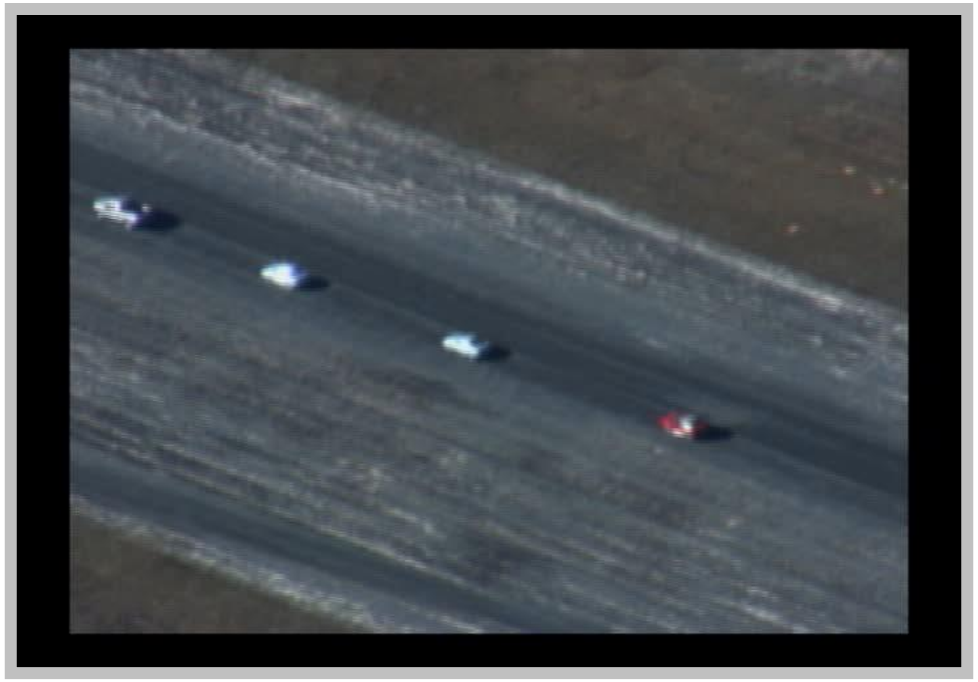
Cross-view action synthesis



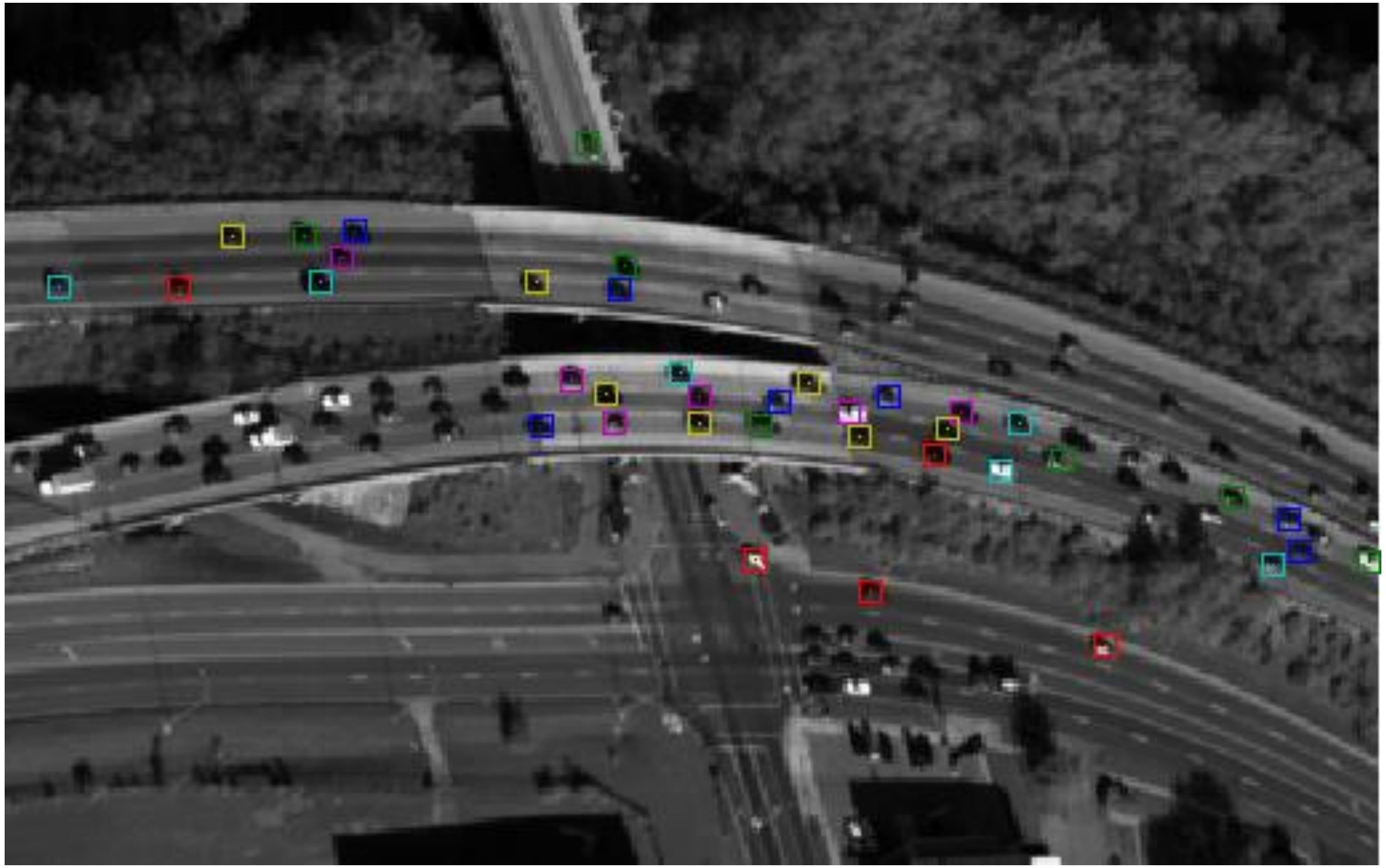
ECCV 2020

Detection in aerial videos

(Object) Tracking

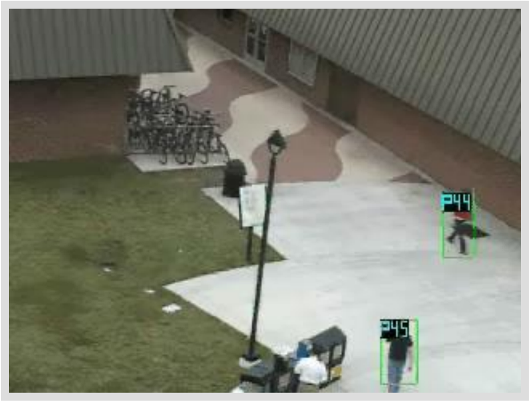


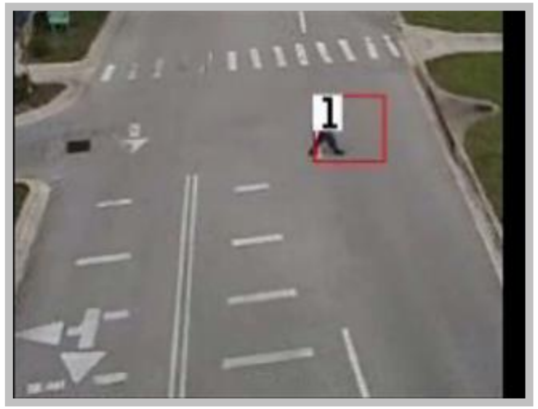
Tracking (multi-object)



Video Security and Monitoring

Object detection → Object tracking → Object categorization and classification → Event or Activities Recognition



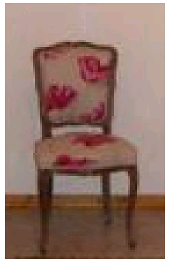


UCF Center for Research in Computer Vision

So…. How do we detect an object in an image?

Naïve approach: Template Matching

This is a chair



Find the chair in this image

Output of correlation

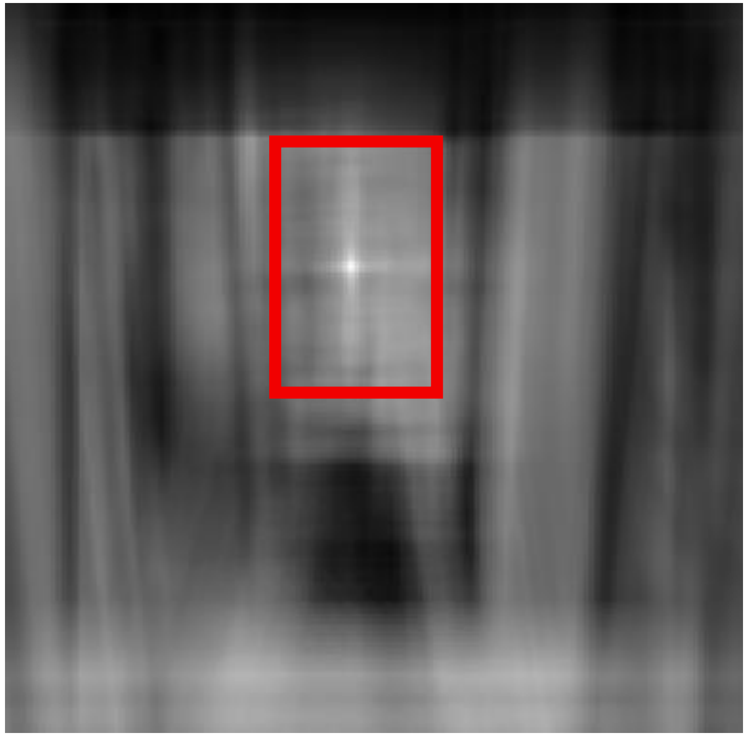
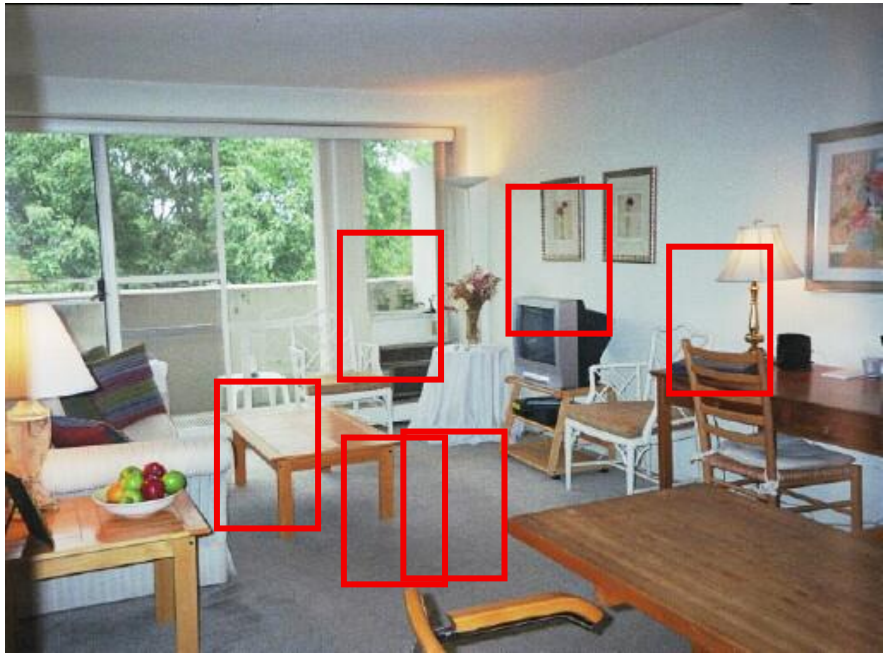
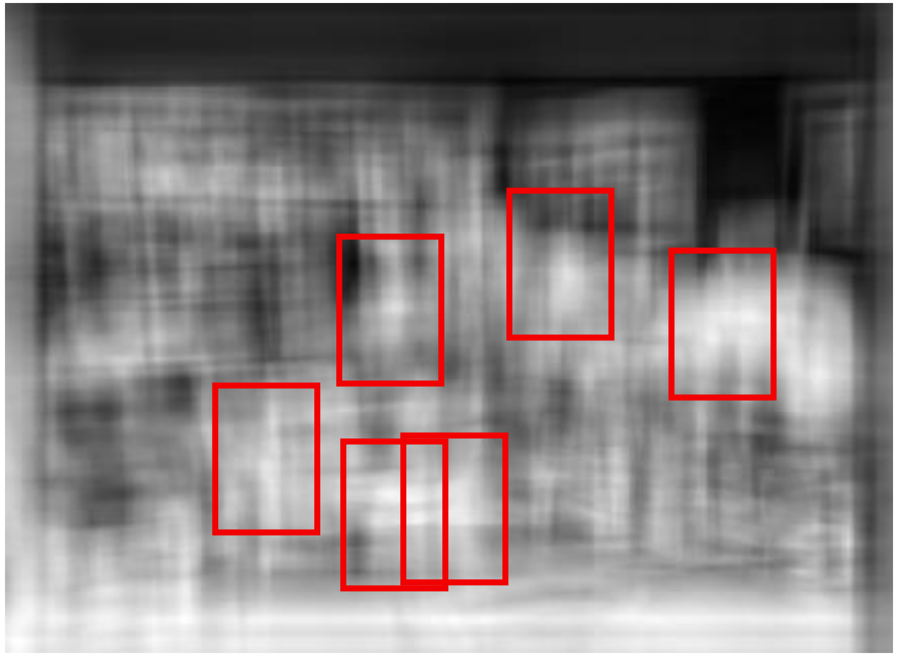


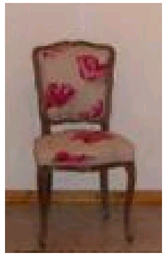
Image source: Antonio Torralba

Template Matching

Find the chair in this image







Epic fail!
Simple template matching is not going to make it

Image source: Antonio Torralba

Idea

- Instead of comparing raw image pixels:
  - First map those pixels into another (more robust) form,
  - And then compare those mapped forms.
  - Finally, select the closest image map (how do you define “closest”? Metrics).
- Features, examples:
  - compute edges
  - compute color histograms
  - gradients
  - HOG
  - SIFT
  - …

"Bag-of-Words" Representation

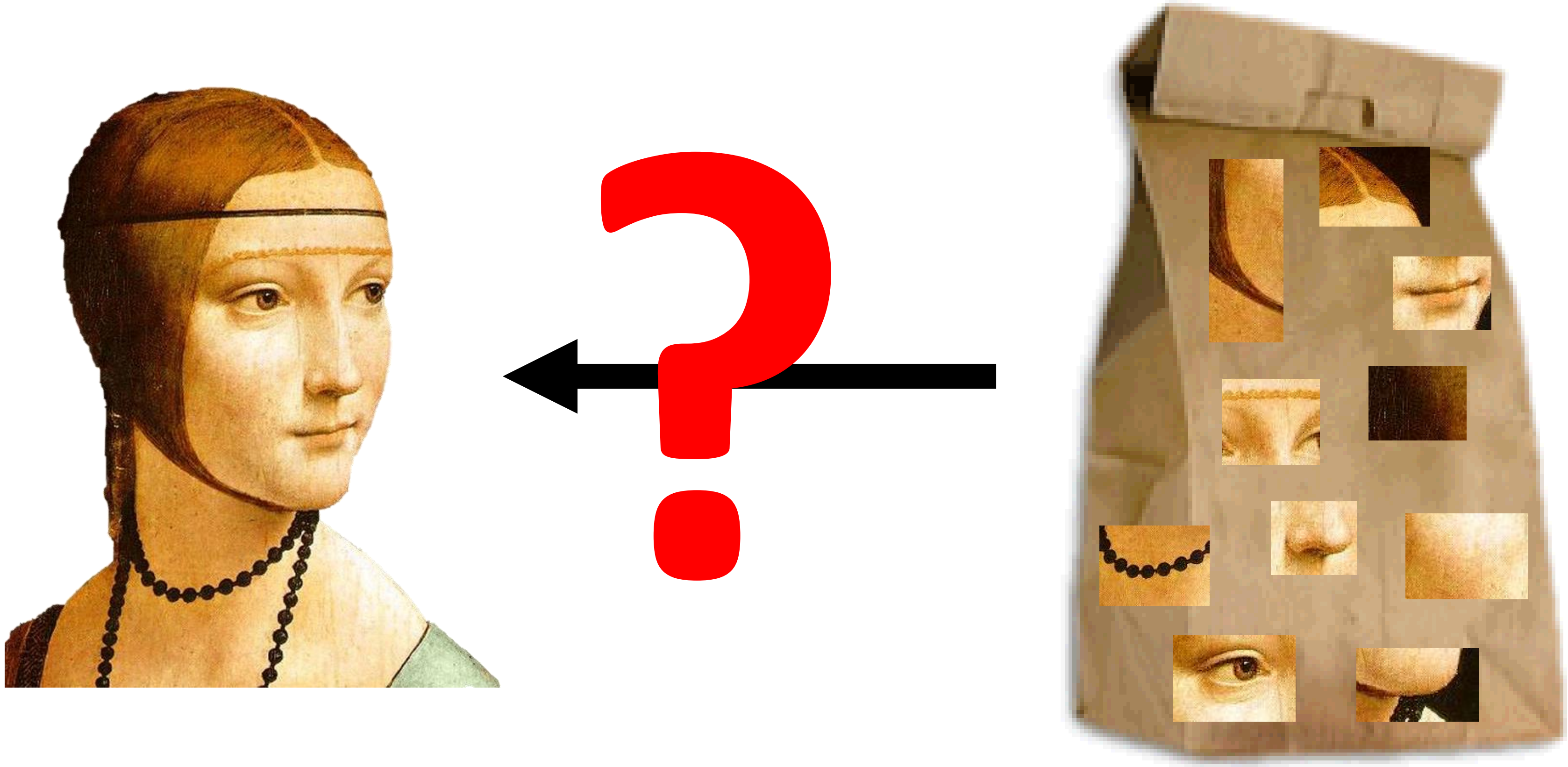


Image source: Antonio Torralba

"Bag-of-Words" (BoW) Histograms

Image

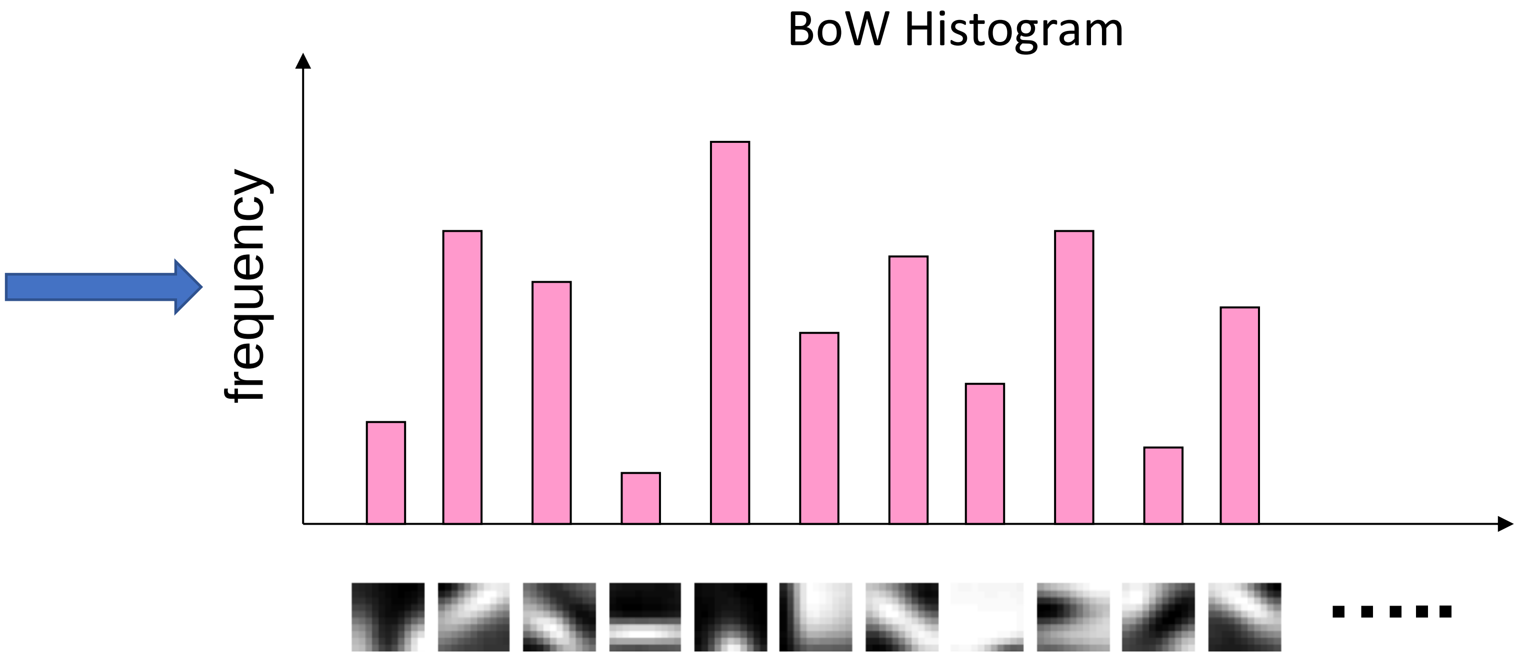


Image Features → BoW Histogram → Classifier → Result

Recipe

BoW Representation

- All have equal probability for bag-of-words methods,
- Location (spatial) information is important but lost.

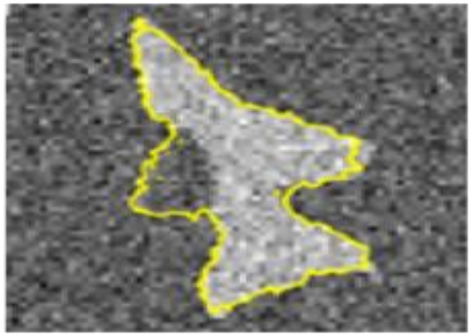
Image source: Antonio Torralba

Computer vs. Human Vision?

- In which task computers are superior to humans?
- In which task humans are superior to computers?

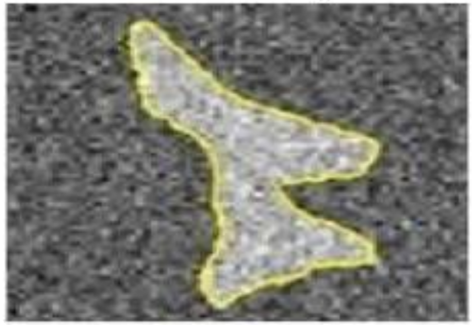
Human > Computer

Recognition, detection, etc … global tasks



Computer > Human

Delineation, local analysis, etc … local tasks



A brief history

50 Years Ago

25 Years ago

15 Years ago

5 years ago

Today

https://youtu.be/PINeQV0LH6k

Is it perfect?

Stop Sign

Speed 60

Questions?

Sources for this lecture include materials from works by Abhijit Mahalanobis, Sedat Ozer, Ulas Bagci, Mubarak Shah, and Antonio Torralba